

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **SIH26160**
- Problem Statement Title – **AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework**
- Theme – **Blockchain & Cybersecurity**
- PS Category – **Software**
- Team ID – **151087**
- Team Name – **Sudo Spandr (Registered on portal)**



Idea: IPsec X-Ray — check how safe an IPsec VPN is, just from its traffic

IPsec X-Ray — AI-Powered IPsec VPN Protocol Analyzer

PROPOSED SOLUTION • the idea • how it solves the problem • what is new



WHAT

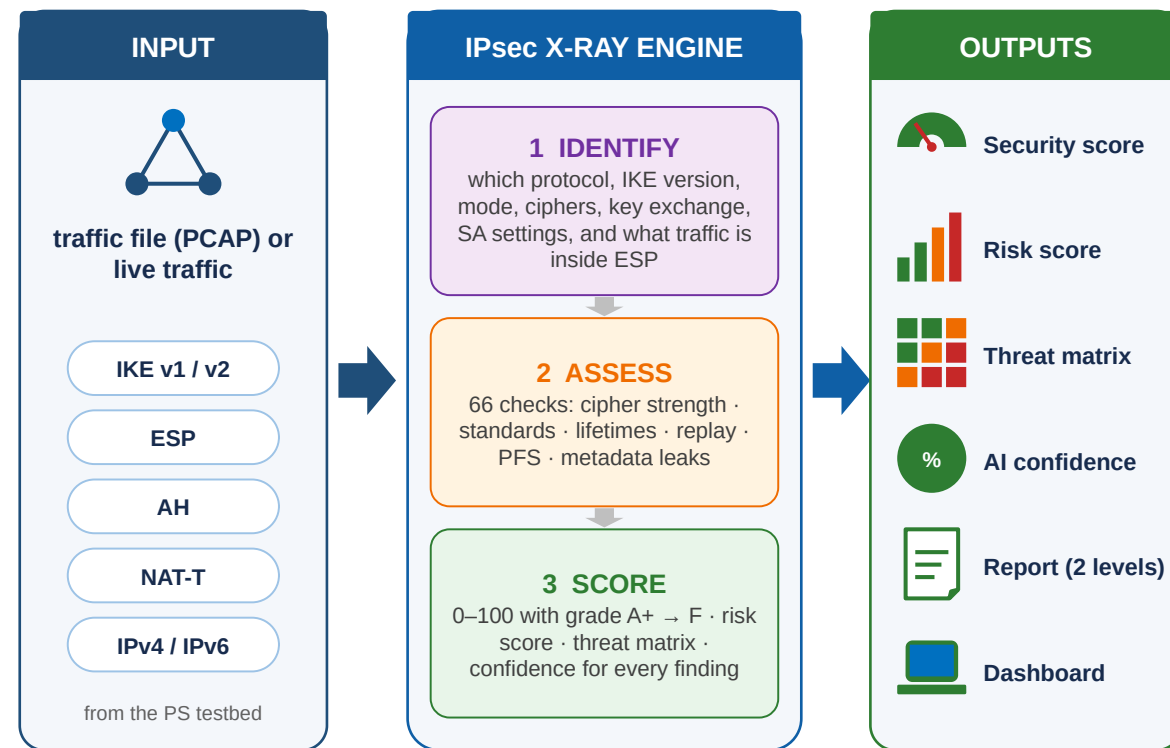
A tool that listens to VPN traffic (IKE / ESP / AH) and tells how safe the tunnel is — no keys, no software on the routers.

HOW

It reads the open parts of the traffic, and uses AI on packet sizes and timing to find the hidden parts: cipher, mode, PFS, and what kind of data is inside (call, chat, mail, web, video).

WHY NOW

Old, breakable settings (IKEv1, 1024-bit DH, no PFS) are still common — and today nobody can see what a tunnel really uses without logging into every device.



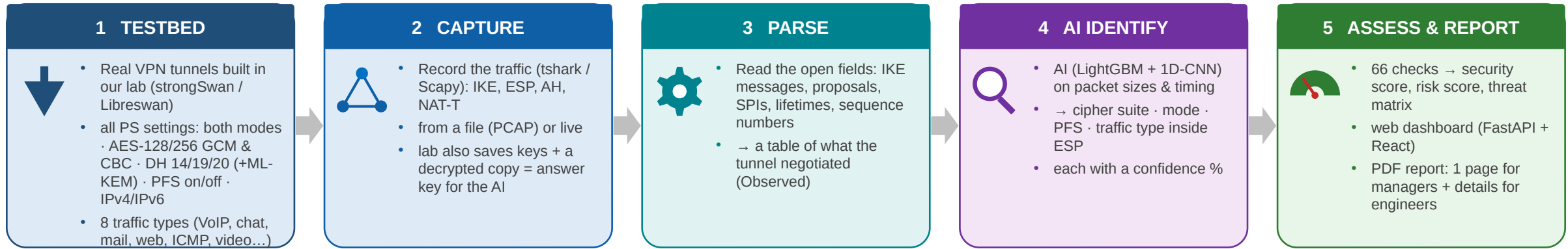
PASSIVE • NO KEYS • NO AGENTS • NO ROUTER ACCESS

WHAT IS NEW

- Sees inside ESP without decrypting it — packet sizes + AI reveal the cipher, mode, PFS and traffic type.
- Never guesses — every result is marked Observed / Inferred / Unknown with a confidence %, and names the standard it checks.

TECHNICAL APPROACH

Technologies used • Step-by-step process • Working prototype

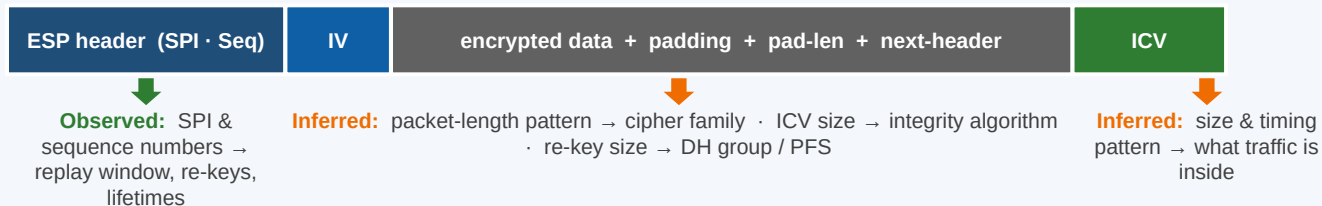


OBSERVED — read directly from the open IKE / ESP headers

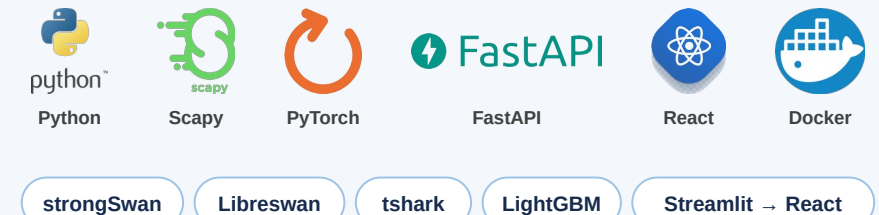
INFERRED — AI + confidence %

SCORED — against the standards

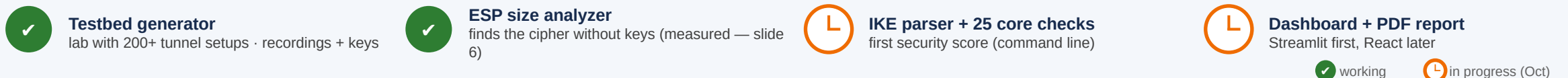
HOW THE AI SEES INSIDE AN ESP PACKET



TECH STACK



WHAT WORKS TODAY



FEASIBILITY AND VIABILITY

Why it can be built • Challenges & risks • How we handle them

WHY IT CAN BE BUILT

OPEN TOOLS & STANDARDS

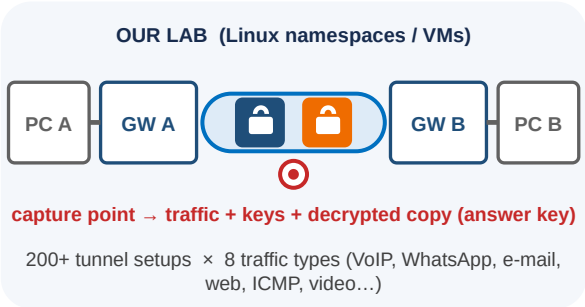
Every PS setting can be built with free tools (strongSwan / Libreswan). Checks follow RFC 8221 / 8247 / 9395, NIST SP 800-77r1 and CNSA 2.0.

OUR OWN LAB = ANSWER KEY

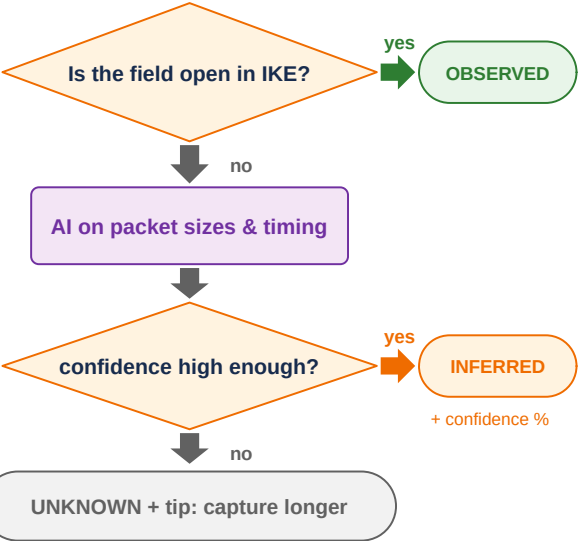
For 200+ tunnel setups we keep the traffic, the keys and the decrypted copy — so the AI is trained and tested on known answers.

AI ONLY WHERE NEEDED

Open fields are read directly. AI is used only for hidden fields, with a confidence %. Low confidence → 'Unknown'.



HOW EACH FIELD IS DECIDED



- OCT**

parser · 25 checks · first score
- NOV**

traffic-type AI · dashboard · dataset
- DEC**

polish · demo video · finale

CHALLENGES → HOW WE HANDLE THEM

Challenge / risk	Our answer
AES-128 and AES-256 look the same on the wire	Say 'AES-GCM, key size unknown' unless IKE shows it — never guess
PFS can be seen only when the tunnel re-keys	Mark 'PFS: Unknown' and tell the user how long to capture
Different vendors pad packets differently	Calibrate per vendor in our lab; lower the confidence if patterns disagree
AI may be over-confident	Calibrated probabilities; below the threshold → 'Unknown'
TLS / QUIC inside the tunnel hides the app	Report 'encrypted web' with lower confidence — no over-claiming
Lab has only strongSwan / Libreswan	Add Cisco / Fortinet / Windows captures in Nov; unknown vendor → 'Unknown'

- STANDARDS**
- RFC 8221
- RFC 8247
- RFC 9395
- NIST 800-77
- CNSA 2.0

Low cost, easy to scale: runs on a normal laptop or a SOC server, 100 % open-source, works offline, no vendor lock-in. New ciphers or vendors are added as simple YAML rules, not code. The checks list and the dataset will be published.

IMPACT AND BENEFITS

Who gains from it • Benefits — security, cost, strategy

SAMPLE RESULT • lab tunnel #37



Grade D — weak

- IKEv1 aggressive mode — deprecated
- DH group 2 (1024-bit) — breakable
- PFS — not seen yet (Unknown)
- AES-CBC + HMAC-SHA2-256 — OK

AI confidence 0.91 • 3 fixes suggested

WHO BENEFITS



NTRO / CERT-In & auditors — check any VPN from a traffic recording; no vendor login needed.



Power, banking, telecom, defence SOCs — watch the crypto health of every tunnel, all the time.



Compliance teams — a clear list of gaps vs RFC / NIST / CNSA, with what to fix first.



Investigators & researchers — learn what kind of traffic a tunnel carries + an open dataset.

BENEFITS



SECURITY

Weak tunnels (IKEv1, DES / 3DES, 1024-bit DH, no PFS) are found — with a score and a fix list.



COST & TIME

Hours of manual checking → minutes per tunnel. Nothing to install, no downtime, free tools.



FUTURE-READY

Shows if a tunnel is ready for post-quantum crypto (ML-KEM) — against 'store now, break later'.



MADE IN INDIA

An Indian, vendor-neutral tool for national and critical networks.

days → minutes

time to audit one tunnel

0 keys · 0 agents

nothing to install

66 checks · 12 threats

score 0–100 · grade A+ → F

WHY IT MATTERS

- **Logjam (2015)**: with one pre-computed 1024-bit DH prime, a nation-state could quietly decrypt about 66 % of IPsec VPN servers.
- **IKEv1 was retired in 2023 (RFC 9395)** — but old tunnels still run, and traffic recorded today can be broken later.

× **Manual check today**: config files and vendor logins — trusts what is written, not what really runs.

✓ **IPsec X-Ray**: looks at the real traffic — what was really negotiated, what is weak, and how sure we are.

RESEARCH AND REFERENCES

Sources and links behind this idea

STANDARDS WE CHECK AGAINST

- [1] RFC 4303 — IP Encapsulating Security Payload (ESP) rfc-editor.org/rfc/rfc4303
- [2] RFC 7296 — Internet Key Exchange Protocol v2 (IKEv2) rfc-editor.org/rfc/rfc7296
- [3] RFC 8221 — Cryptographic algorithm requirements for ESP & AH rfc-editor.org/rfc/rfc8221
- [4] RFC 8247 — Algorithm implementation requirements for IKEv2 rfc-editor.org/rfc/rfc8247
- [5] RFC 9395 — Deprecation of IKEv1 and obsoleted algorithms rfc-editor.org/rfc/rfc9395
- [6] NIST SP 800-77 Rev. 1 — Guide to IPsec VPNs csrc.nist.gov/pubs/sp/800/77/r1/final
- [7] NSA CNSA 2.0 — DH 16/20, AES-256, SHA-384, ML-KEM (post-quantum) nsa.gov/Press-Room/Cybersecurity-Advisories-Guidance/

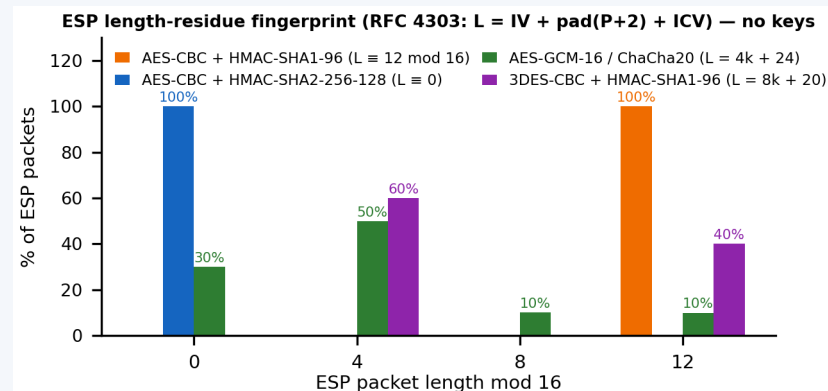
RESEARCH, DATA & TOOLS

- [8] Adrian et al., “Imperfect Forward Secrecy” (Logjam), ACM CCS 2015 weakdh.org
- [9] Felsch et al., “The Dangers of Key Reuse: Practical Attacks on IPsec IKE”, USENIX Security 2018 [usenix.org/conference/usenixsecurity18/presentation/felsch](https://usenixsecurity18/presentation/felsch)
- [10] Bhargavan & Leurent, Sweet32 — 64-bit block-cipher (3DES) collision attacks sweet32.info
- [11] ISCX VPN-nonVPN 2016 dataset (UNB CIC) — encrypted-traffic classification baseline unb.ca/cic/datasets/vpn.html
- [12] ET-BERT — transformer for encrypted-traffic classification (arXiv 2202.06335) arxiv.org/abs/2202.06335
- [13] IETF draft-ietf-ipsecme-ikev2-mlkem + strongSwan 6.0 (ML-KEM in IKEv2) — PQC testbed tunnels datatracker.ietf.org/doc/draft-ietf-ipsecme-ikev2-mlkem/

WHAT WE WILL DELIVER • PS DELIVERABLES

- ✓ Prototype analyzer + AI engine
- ✓ IPsec testbed generator
- ✓ Interactive dashboard
- ✓ Executive + technical report
- ✓ Demo video + documentation
- ✓ Labelled IPsec dataset (open)

OUR FIRST RESULT • LAB MEASUREMENT



Packet-length patterns alone tell the cipher suites apart — no key needed. This is the base of the AI step.